FROM YOU

YOUR SUGGESTIONS AND THINGS YOU WANTED TO KNOW!

THE E-ZINE

Thanks for the e-zine issues! It was exciting to see various topics covered in your magazine, which makes it one of the most informative and advanced leading electronic technical magazines in India, no doubt. Keep it up! It's great to receive the e-zine; I am sharing it with my professional colleagues. Thanks!

Anthony D'Sa
Consultant, Control & Automation Systems

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BATTERIES PROTECTION

Recently I read the 'Protection Mechanisms For Batteries And Their Charging Systems' article in January issue. It's really useful. I have learned a lot. Thanks to EFY and the author!

Prakash U.M.

EFY. Thanks for the feedback!

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TOY CAR

I am interested in the 'Remote Controlled Toy Car Demystified' DIY article published in October 2021 issue. Can I get the PCB and other electronic items for this project?

Senthil

EFY. Please visit www.kitsnspares.com or write to *info@kitsnspares.com* for the project kit, including the PCB, electronic components, and project manual.

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APPLIANCES PROTECTION CIRCUIT

I wish to draw your attention to an error in the circuit diagram of 'Circuit To Protect Appliances From High/Low Voltages' DIY article published in February issue. The diode across the relay coil is wrongly drawn. Its anode should be connected to the collector of transistor and cathode to the + ve line.

Ramprakash Venugopalan

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HUMAN-PRESENCE SENSOR

There seems to be a typographical error in 'Human-presence Sensor' article under the Truly Innovative Tech section in January issue. In the first line, shouldn't "Ultra-sideband" be Ultra-Wideband?

Prakash U.M.

EFY. Thanks for pointing out the mistakes above! We highly appreciate your quick alerts, which would help the other readers.

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BATTERY CHARGER

My 48V, 5A lead-acid battery charger is not working. So, I have following questions:

1. How can I repair my battery charger?
2. What are 25NF60N and 330MF250V codes in the charger circuit?
3. If the two MOSFETs and capacitors are replaced, would the MOSFETs blow off?
4. Would the charger function, if I replaced these components?
5. How to avoid blowing off components during soldering?

Roy
Kolkata

EFY. 1. While some battery chargers can be easily repaired, the chargers with SMD components are not so easy to repair. However, to repair, first you need to identify the exact problem. Check for any loose connection, short-circuit, faulty components, or a blown fuse (if used) in the circuit. Based on the problem you would need the required tools and parts to repair it.

2. The codes you are referring to seem be the part names of the components. Here, 25NF60N and 330MF250V could be an N-channel MOSFET and electrolytic capacitor, respectively.

3. It is not clear why you would like to replace MOSFETs and capacitors. If these are damaged, first try to find the root cause. If you replace them with wrong components or polarities, the MOSFETs could blow off or may not work.

4. After rectifying the root cause, if you replace the faulty components with appropriate components, it may function again, provided you have not damaged the PCB tracks and other associated components in the PCB.

5. SMD components often get damaged during soldering, but such cases are rare with through-hole components. One of the most common causes of damaging the SMD components is excessive heating with hot air gun. If you apply excessive heat to an SMD component, it may get damaged and even blow off or explode. To prevent that, first apply proper solder paste on the pads, place the components on their exact location on the PCB and then apply just the required amount of heat. Use a fine-tip tweezer to hold the component.

You need to look at typical thermal profiles of each component in the datasheet before soldering. Apply heat or set the temperature of the soldering machine accordingly. Cooling fan may be required, but do not bring the fan too close until solder melts. For lead/tin solder, you can set temperature between 260°C and 280°C and then heat a joint for about 15 to 20 seconds from the top, until solder melts and the component settles.

Sometimes improper soldering leads to solder bridging, where extra solder application shorts two points by mistake. When power is switched on, a solder bridge can burn or blow up a component and can even burn out a PCB track!

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